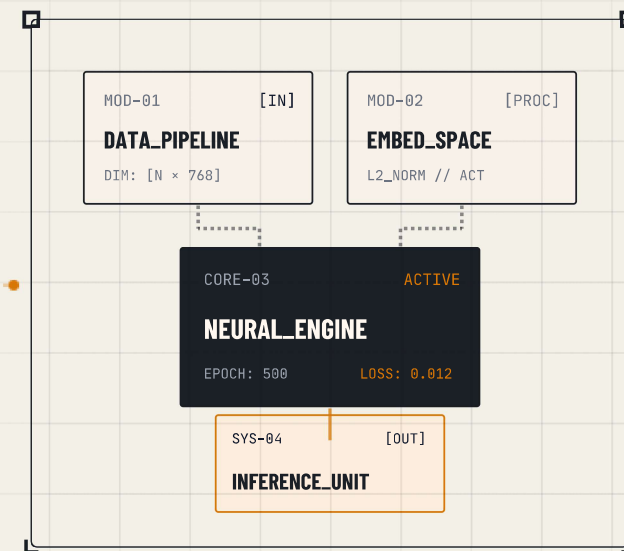


● ENGINEERING MANUAL

THE WEEKEND HACKER'S GUIDE TO MACHINE LEARNING

AUTHOR / PRESENTER

Ayush Lahiri



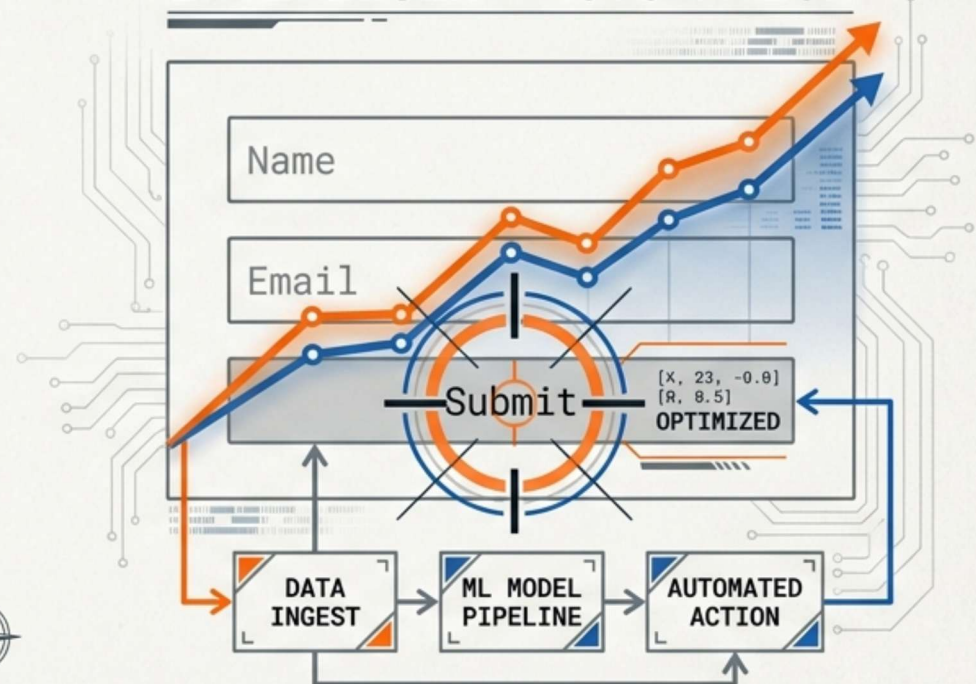
WHY MACHINE LEARNING WINS HACKATHONS

Judges see hundreds of generic web apps. Machine Learning cuts through the noise by solving specific real-world problems with intelligent automation. You don't need a PhD; you just need to wire the right modular blocks together.

GENERIC CRUD APP

A simple web form with three input fields: 'Name', 'Email', and a 'Submit' button. The form is enclosed in a rectangular box.

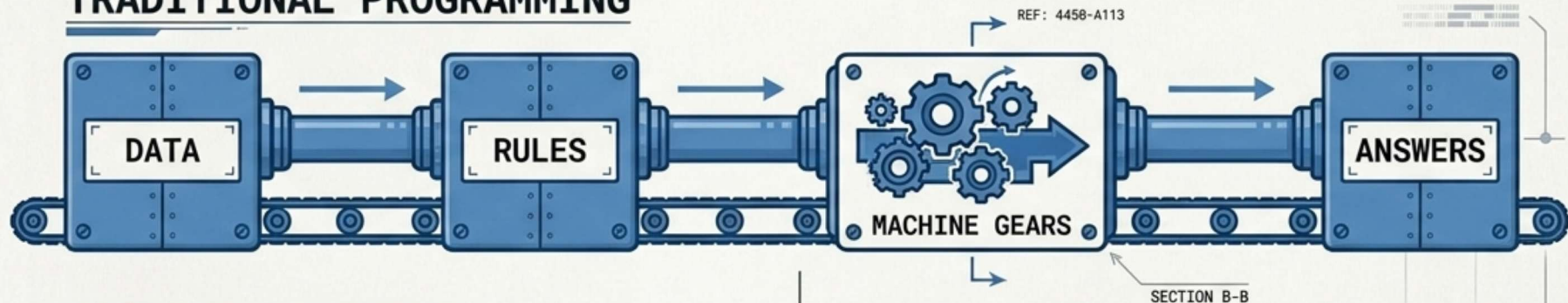
INTELLIGENT AUTOMATION



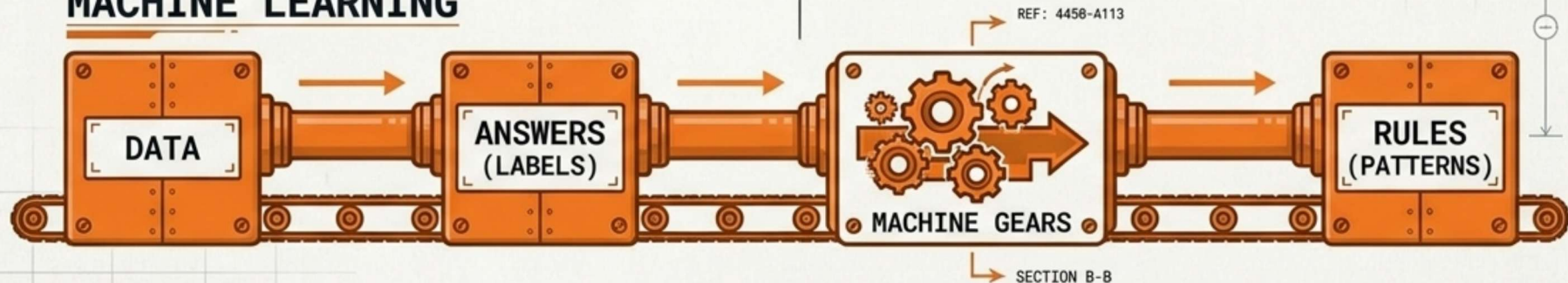
THE CORE PARADIGM SHIFT

Traditional programming relies on explicit if-then rules. Machine Learning flips the equation: feed the computer empirical data and correct answers, and the algorithm implicitly learns the underlying patterns.

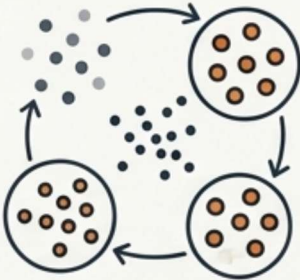
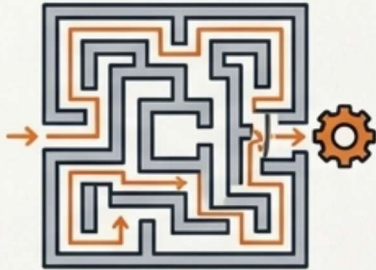
TRADITIONAL PROGRAMMING



MACHINE LEARNING



THE RULES OF ENGAGEMENT: THREE ML PARADIGMS

SUPERVISED LEARNING	UNSUPERVISED LEARNING	REINFORCEMENT LEARNING
GOAL: Predict Outcomes	GOAL: Find Clusters & Structures	GOAL: Trial & Error Optimization
DATA TYPE: Labeled Data	DATA TYPE: Unlabeled Data	DATA TYPE: Reward/Penalty Signals
EXAMPLE: Spam vs. Not Spam	EXAMPLE: Customer Segmentation	EXAMPLE: Game Bots
		

TACTICAL NOTE: FOR A 24-HOUR HACKATHON, SPEND 90% OF YOUR TIME IN SUPERVISED LEARNING.



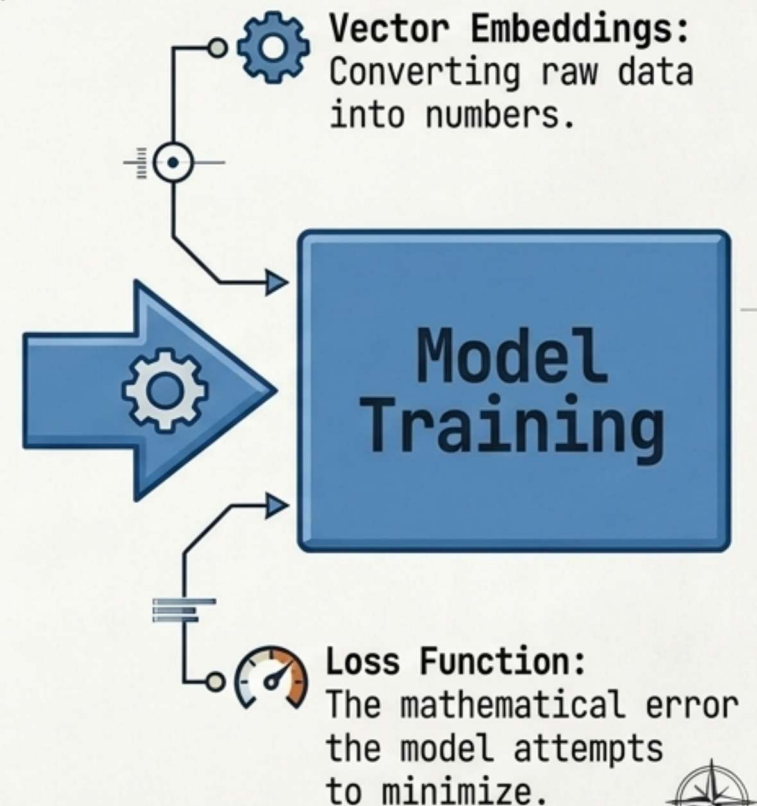
THE ANATOMY OF A DATASET

Features
(The Measurable Inputs)

Label
(The Target Outcome)



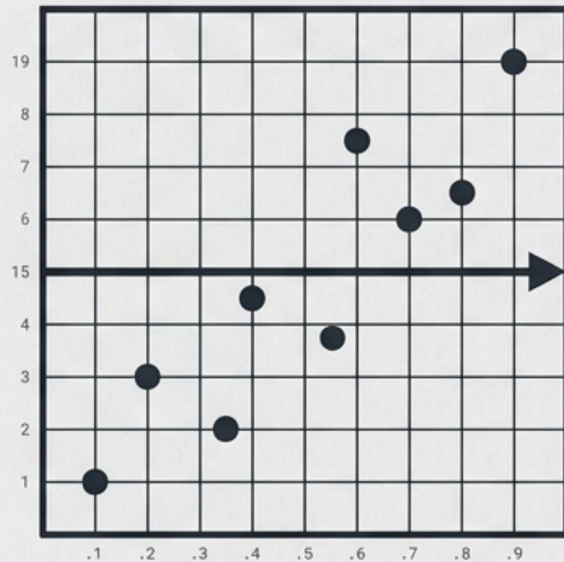
Sq Ft	Bedrooms	Age	Price
2500	3	15	\$500k
1800	2	8	\$380k
3200	4	20	\$650k
1200	2	30	\$290k
2100	3	5	\$420k



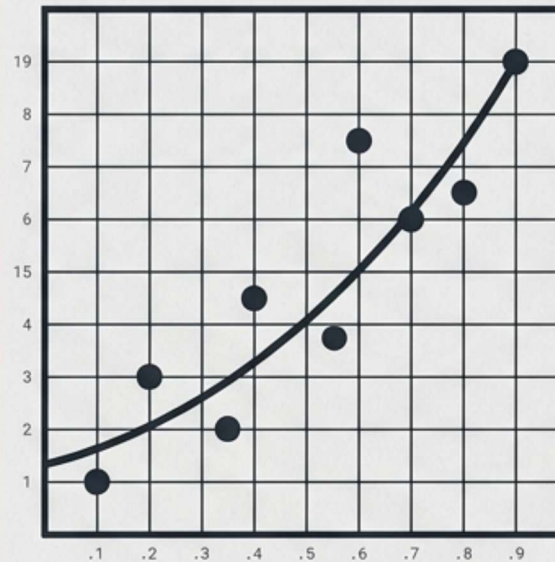
THE GENERALIZATION TEST

We use Train/Test splits to hide 20% of our data during training. If the model fails on the hidden test data, it memorized the noise instead of learning the pattern.

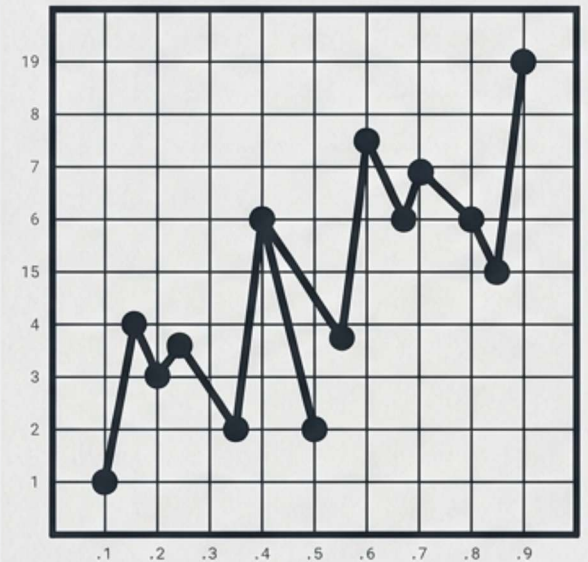
**UNDERFITTING
(TOO SIMPLE)**



**PERFECT FIT
(GENERALIZES WELL)**

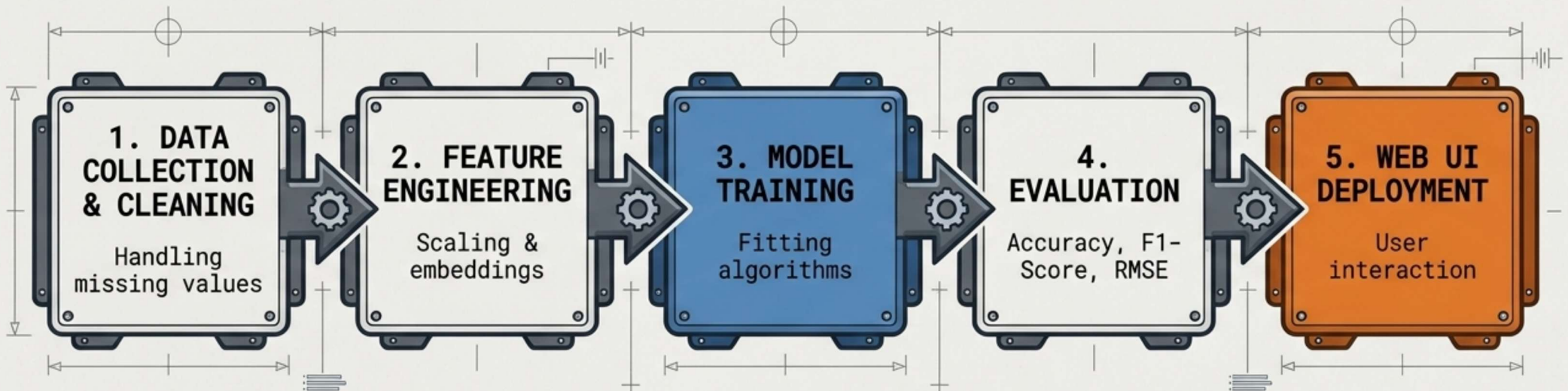


**OVERFITTING
(MEMORIZED NOISE)**



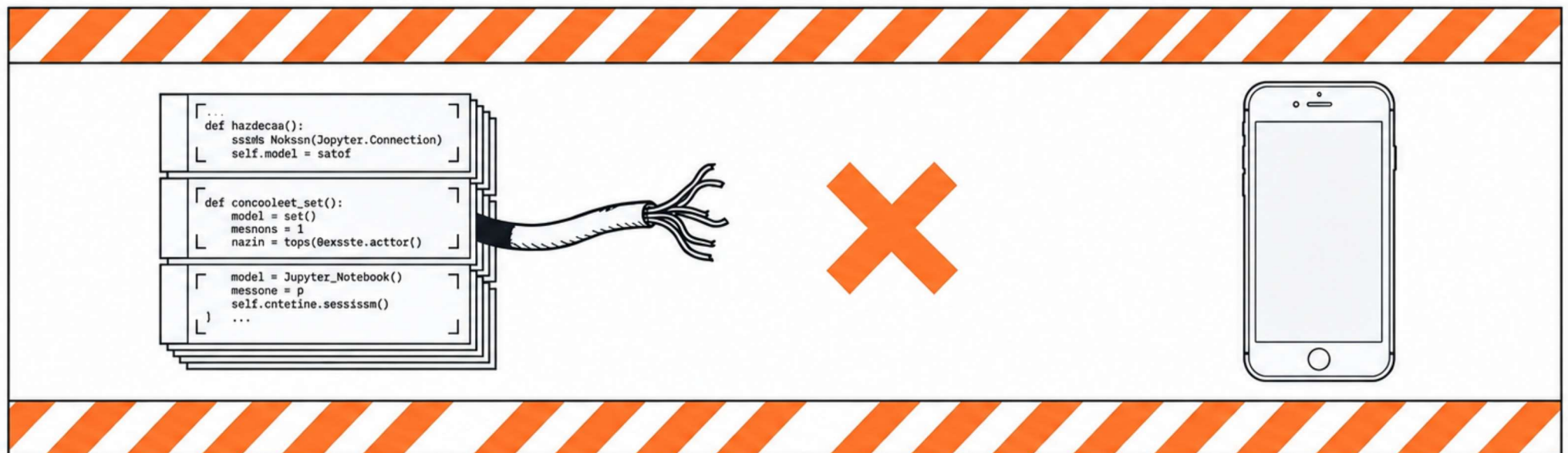
THE MASTER BLUEPRINT: END-TO-END WORKFLOW

This is the exact sequence you will execute. You cannot skip a block, and the model is only one piece of the machine.



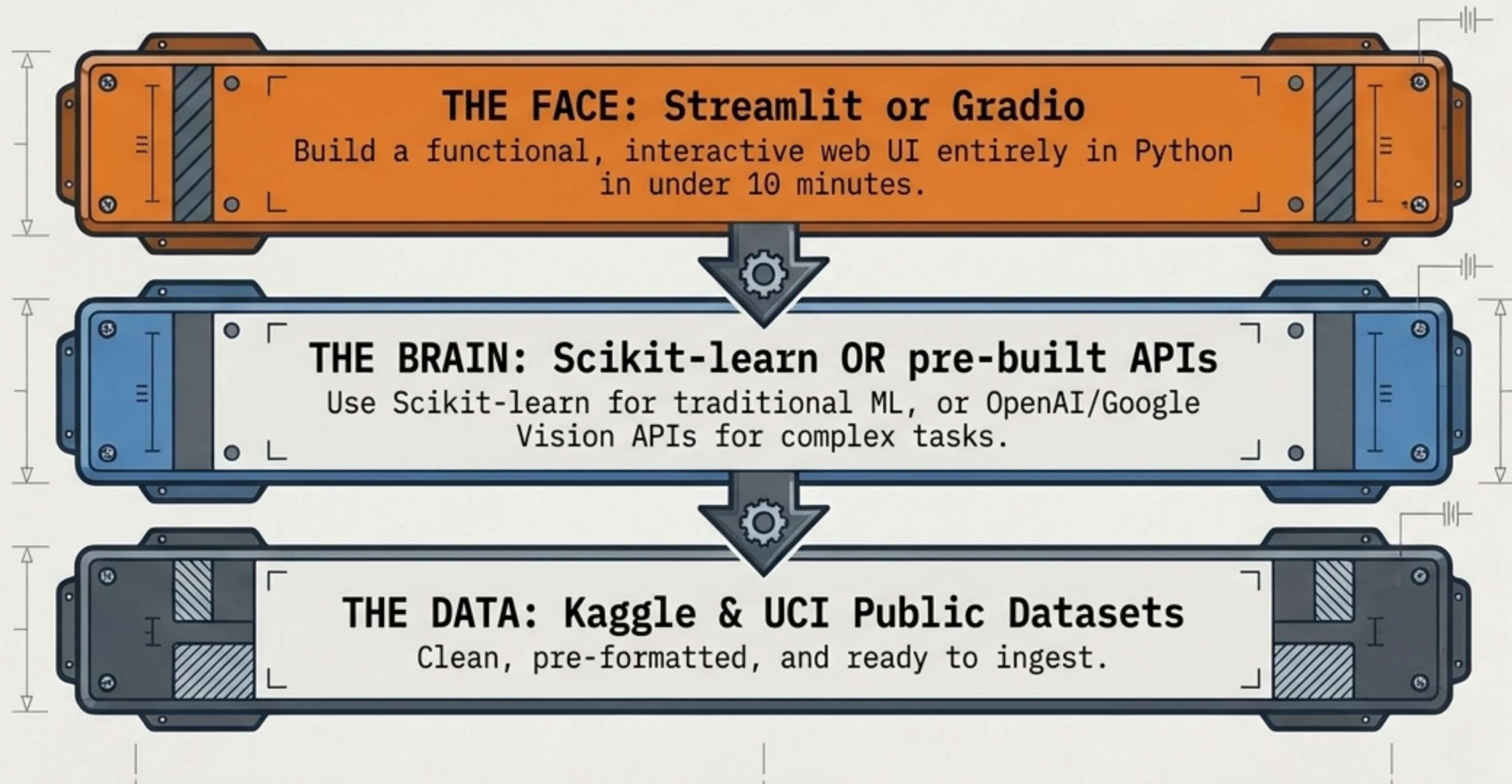
A Model in a Jupyter Notebook is NOT a Product.

The harshest hackathon reality check: Judges do not care about your 99% accuracy score if they cannot click a button and interact with it themselves. The ML Model is only 10% of a winning project; the interface is the other 90%.



THE HIGH-SPEED TECH STACK

Do not build a React front-end from scratch if your core value is an ML feature. Optimize for speed.

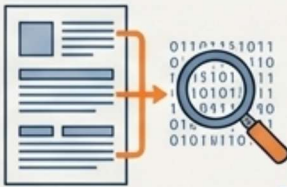


MISSION BRIEFINGS: TOP 5 BEGINNER PROJECTS

High impact. Narrow scope. Public data readily available.

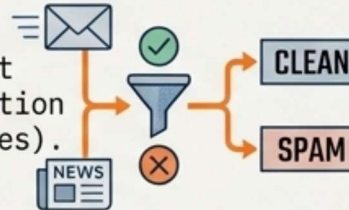
AI RESUME REVIEWER

NLP text classification & similarity scoring.



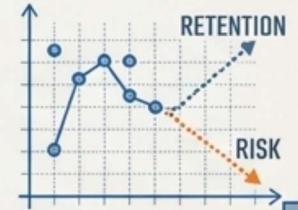
SPAM/FAKE NEWS CLASSIFIER

Binary text classification (Naive Bayes).



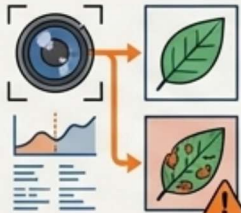
CUSTOMER CHURN PREDICTOR

Tabular data regression for retention risk.

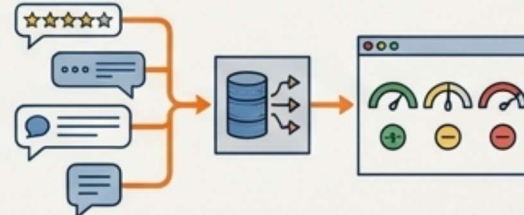


PLANT DISEASE DETECTOR

Computer vision (Image classification).

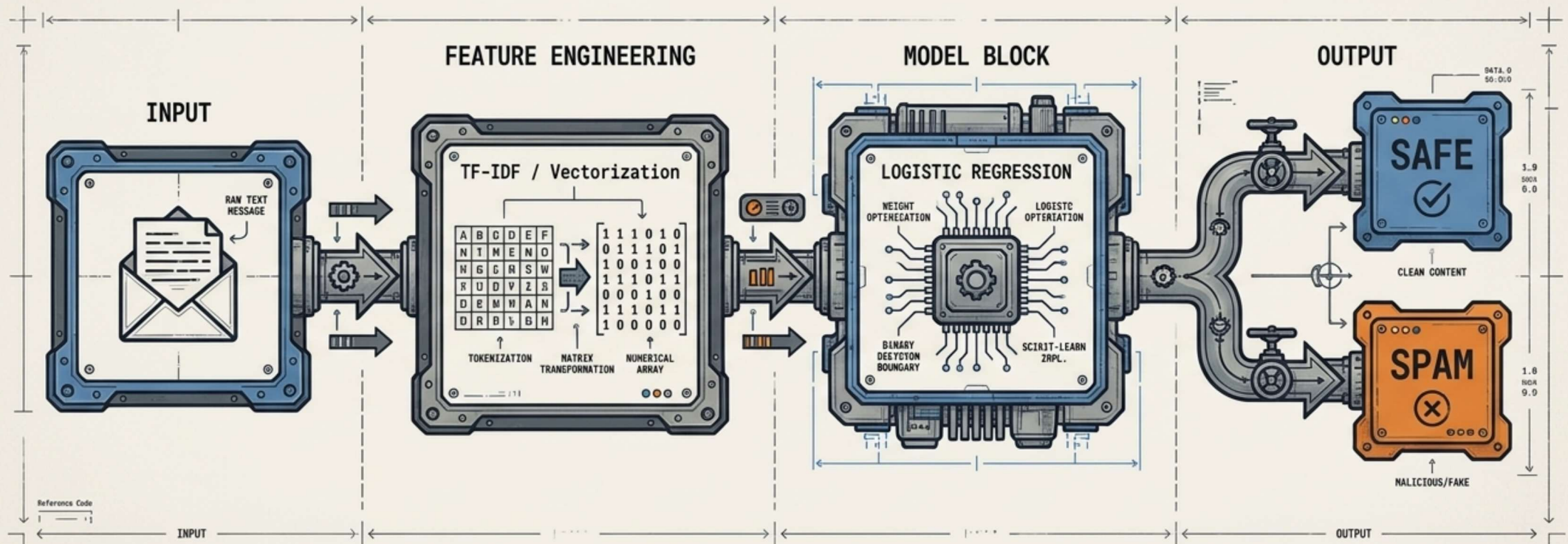


SENTIMENT DASHBOARD



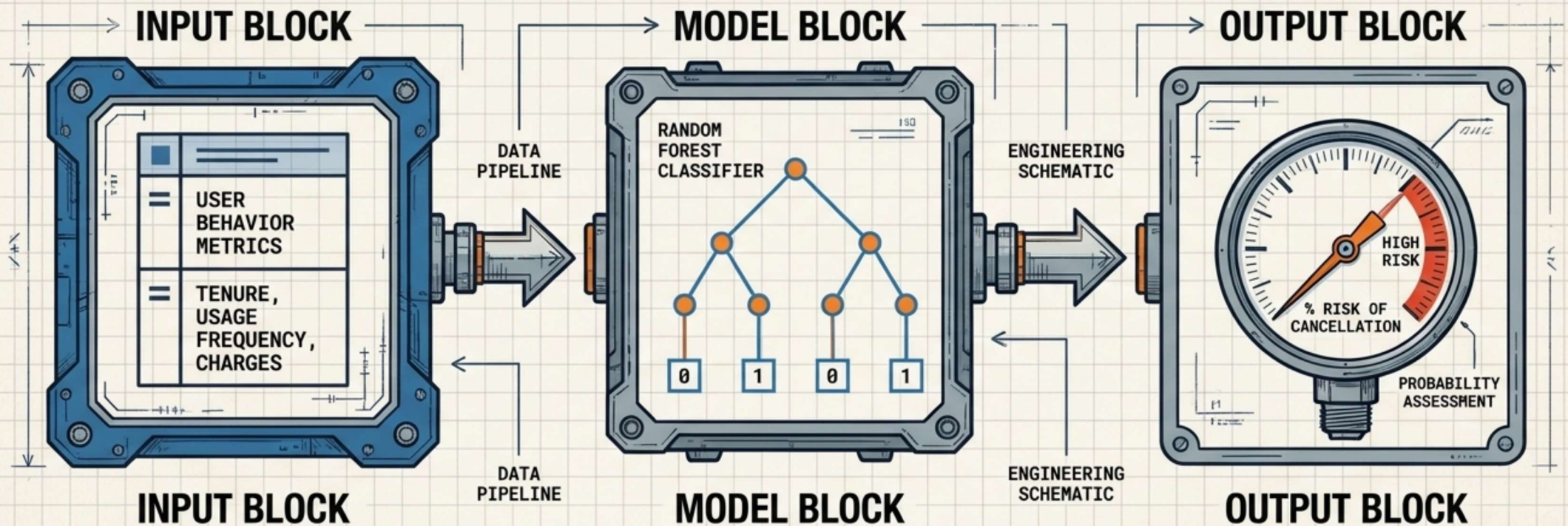
PROJECT ARCHITECTURE: TEXT CLASSIFICATION

The classic binary task: Fake News or Spam Classifier. Perfect for learning how to handle raw, unstructured text and evaluating false positives.



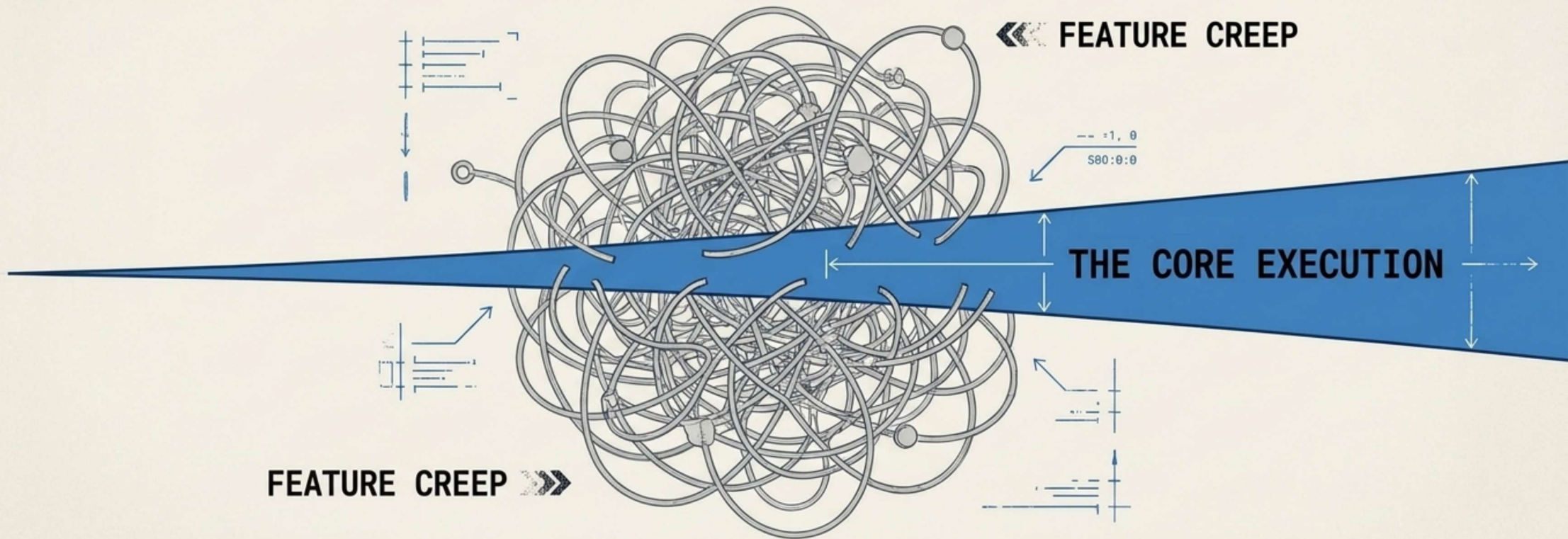
PROJECT ARCHITECTURE: BEHAVIORAL PREDICTION

Hackathon judges reward projects that solve commercial problems. Flagging an at-risk customer early allows retention teams to intervene.



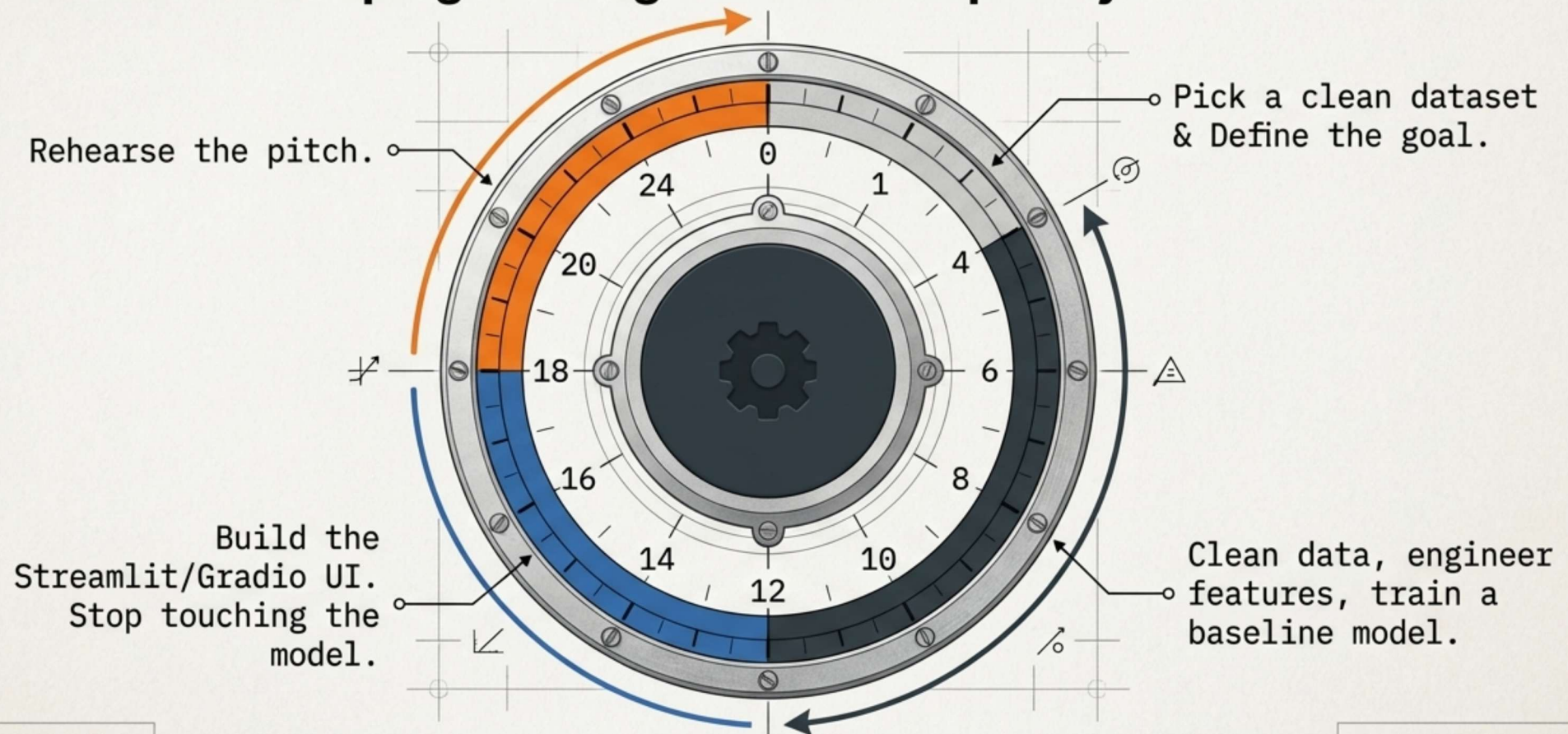
SCOPE TO ONE WORKING FEATURE

Do not build a platform. Do not build a suite of tools. A live demo with one polished, working core feature beats a demo with five half-broken features. Cut everything that isn't required to prove the ML concept.



THE 24-HOUR EXECUTION TIMELINE

The most common point of failure is tweaking the model until hour 23. Execution and scoping beat algorithmic complexity.



THE FINAL BOSS: THE PITCH

The technology is just a tool. The story you tell the judges is what wins the weekend. Go build it.

